PROJECT REPORT · INFORMATION RETRIEVAL

ResearchOS: Hybrid Neural Search for AI Research Papers

A retrieval system that finds research papers by what they do, and explains the citations inside them. Hybrid lexical and dense retrieval, fused by reciprocal rank fusion and reranked by a fine-tuned cross-encoder, evaluated by a six-configuration ablation on a public benchmark and deployed as a public service.

Benchmark	Headline result	Hardware	Status
SciFact (BEIR), 300 queries	0.760 nDCG@10, +10.8% vs BM25	RTX 4060, 8 GB VRAM	Deployed, public, tested

1 The problem

Searching papers by keyword fails in a specific way: you rarely remember the words a paper used, you remember what it *did*. Two retrieval families address this and each fails where the other succeeds. **BM25** scores documents by weighted word overlap — fast, untrained, exact on rare technical terms, but unable to match *car* to *automobile*. **Dense retrieval** embeds text into vectors where proximity means semantic similarity, bridging vocabulary gaps but blurring precise terms. The system runs both, fuses the rankings, then re-scores the shortlist with a cross-encoder. Each of those three decisions was measured rather than assumed.

2 Architecture

01 **Parallel retrieval**
BM25 over a stemmed index and dense retrieval over a FAISS flat index run independently against the same passages. Flat rather than approximate: at 40k passages exact search is already under 10 ms, so an ANN index would trade recall for speed nobody needs.
bm25s + PyStemmer · faiss-cpu · bge-small-en-v1.5 (33M)

02 **Reciprocal Rank Fusion**
The rankings merge by *position*, not score — BM25 is unbounded and cosine similarity is bounded, so there is no principled way to add them. Each document scores 1/(k+rank) summed across lists, so documents both retrievers rank highly rise.
k = 60 · top-60 candidates (GPU) / top-30 (CPU deployment)

03 **Cross-encoder reranking**
The shortlist is re-scored by a model reading query and passage concatenated, so attention runs across both. Far more accurate than comparing independent embeddings, and far too slow to run over the full corpus — hence its place after a cheap first pass.
bge-reranker-base (278M), fine-tuned · max_length 384

04 **Confidence gate and generation**
The mean of the top-5 reranker scores decides whether the corpus covers the query at all. Below 0.35 the system searches arXiv live instead of returning confident nonsense. Passing results go to an LLM constrained to the retrieved passages only.
mean-of-top-5 threshold · Groq, openai/gpt-oss-20b

DESIGN DETAIL WORTH DEFENDING
The gate uses the **mean** of the top 5, not the maximum. An earlier version used the max and confidently returned a bone-densitometry paper for a knee-osteoarthritis query: max 0.670 looked convincing, but the mean was 0.190. One lucky partial match spikes a maximum; it cannot lift a mean. Measured separation: in-corpus 0.67–0.996, out-of-corpus 0.001–0.19.

3 Data

Two datasets, and keeping them separate matters for every claim here. **SciFact** (BEIR) has human relevance judgments, so every metric in section 4 comes from it — 300 held-out queries, seed 42, scored with the standard TREC evaluation library rather than bespoke metric code. The **deployed arXiv corpus** holds 8,330 papers and 40,379 passages across cs.LG, cs.AI, cs.CL, cs.CV, cs.NE, cs.RO and stat.ML. Papers are chunked at 220 words with 40 words of overlap, section-aware, references dropped: a 30-page paper contains one relevant paragraph, and embedding the whole thing averages the signal away.

SCOPE LIMIT, STATED UP FRONT

The arXiv corpus has **no human relevance labels**, so it cannot produce a trustworthy nDCG figure and none is claimed for it. The 0.760 is SciFact, not production. The deployed system is evaluated by integration tests and click/vote logging instead.

4 Results: the ablation

Six configurations, each adding exactly one component to the one above, so every change is attributable to a single decision.

Configuration	nDCG@10	Recall@10	MAP@10	p50
BM25 only (baseline)	0.6863	0.8187	0.6387	0.68 ms
+ dense, fused with RRF	0.7236	0.8571	0.6756	34 ms
+ off-the-shelf reranker	0.7124	0.8394	0.6670	236 ms
+ fine-tuned, random negatives	0.7583	0.8769	0.7157	214 ms
+ fine-tuned, hard negatives	0.7602	0.8879	0.7152	208 ms
Fine-tuned, ONNX INT8	0.7504	0.8846	0.7038	—

SciFact (BEIR), 300 queries, seed 42. Latency is p50 for the full pipeline on an RTX 4060. Row 3 is shown in red because it scored *below* the row above it.

What each row proves

- **Hybrid retrieval earns its place: +5.4%.** Adding dense retrieval and RRF lifts nDCG@10 from 0.6863 to 0.7236, for roughly 34 ms.
- **An off-the-shelf reranker actively hurt: –1.5%.** Bolting on bge-reranker-base unmodified *dropped* nDCG@10 to 0.7124 — worse than not reranking, while costing 200 ms. The base model was trained on web QA, where relevance means something different than it does for scientific claim verification.
- **Fine-tuning is what makes reranking pay: +6.7%.** The same architecture trained on in-domain data reaches 0.7602, turning a regression into the best configuration.
- **End to end: +10.8% nDCG@10, +8.5% Recall@10, +12.0% MAP@10** over the BM25 baseline. Consistency across all three metrics matters more than any single number.

5 Significance testing and a null result

Rows 4 and 5 differ by 0.0019 nDCG@10. On 300 queries that could easily be noise, so it was tested rather than claimed. **Hard negatives** are passages that look relevant but are not, mined by running retrieval and keeping the top-ranked wrong answers; the theory is that they teach distinctions random negatives cannot.

+0.0019	p = 0.806	[-0.013, +0.017]	40 / 35 / 225
mean difference	permutation test	95% CI, 10k resamples	better / worse / tied

- **Paired bootstrap.** Resampling the 300 queries with replacement 10,000 times gives a 95% interval of [–0.0134, +0.0172]. It contains zero: “hard negatives are slightly worse” is as consistent with the data as “hard negatives help”.
- **Permutation test.** Randomly reassigning which system gets credit per query, 10,000 times, produces a gap this large or larger 80.6% of the time by chance alone.
- **Why paired.** Both systems answer identical queries, so comparison is per-query and the variance from some queries simply being harder cancels out. An unpaired test on the same data would be markedly less sensitive.

The decision: hard-negative mining adds a retrieval pass, a curation step and pipeline complexity, and bought nothing measurable, so it was not shipped as a claimed improvement. The defensible statement is not “hard negatives don’t work” but *“they produced no measurable gain at this scale, on this dataset”* — with 225 of 300 queries tied, the test had very little to work with.

6 Quantization

The reranker was exported to ONNX and quantized to INT8 to test CPU deployment. The comparison that means anything is same-device: INT8 against FP32, both on CPU.

Backend	Device	p50	Model size	Throughput
PyTorch fp16	CUDA	174 ms	—	289 pairs/s
ONNX FP32	CPU	7,790 ms	1,112 MB	6.4 pairs/s
ONNX INT8	CPU	3,879 ms	279 MB	10.3 pairs/s

Reranking latency, 50 queries × 50 candidates. The PyTorch row runs on GPU and is deliberately included to show why it is not comparable.

Against FP32 on the same CPU, INT8 is **2.0× faster and 4.0× smaller**, costing **1.3% nDCG@10** (0.7602 → 0.7504). Comparing ONNX-on-CPU against PyTorch-on-GPU instead would make quantization look catastrophically slow — that is a hardware result, not a quantization result, and any speedup claim has to hold device and batch fixed. INT8 was not deployed: production runs PyTorch on CPU with the candidate set halved to 30 instead.

7 Training the reranker

Loss: MultipleNegativesRankingLoss, a listwise objective — one correct passage against 7 sampled negatives, scored together with cross-entropy over the group. Pointwise binary cross-entropy would ask “is this relevant?” in isolation, which is not the task: the metric rewards ranking the best passage above good-but-worse ones, so training objective and evaluation metric should agree. **Setup:** 384 tokens, batch 8 × 4 gradient accumulation (32 effective), lr 2e-5 cosine with 10% warmup, 3 epochs, fp16, seed 42 — identical across both ablation arms, which differ only in the negatives file.

Fitting 278M parameters into 8 GB. The listwise loss expands each batch item into 8 pairs, so batch size 8 means 64 sequences of 384 tokens per forward pass, and full fine-tuning OOMs immediately. Gradient checkpointing solved it — activations are recomputed during the backward pass rather than stored, costing about 30% throughput and landing at 7.9 GB of 8.0. LoRA r=16 was wired as an automatic fallback on CUDA OOM but never fired. Sequence length, batch size and dataset size were deliberately never reduced: they are pinned by the experiment design, and shrinking them to fit memory would have made the two ablation arms incomparable.

8 The production system

- **Backend** — FastAPI serving retrieval, citation resolution, PDF upload and feedback logging, with a Gradio interface mounted onto the same app so one process serves both humans and the API. Hosted on Hugging Face ZeroGPU.
- **Frontend** — Next.js / TypeScript on Vercel, calling the backend over CORS.
- **Generation** — Groq running gpt-oss-20b, constrained to retrieved passages so citations cannot be fabricated, with every answer carrying the passage IDs it drew on.
- **Verification** — a 13-check integration suite exercising real HTTP endpoints against the live deployment rather than importing modules, so it catches wiring failures (an unregistered route, a stale config) that unit tests miss entirely.

9 Limitations

- **The second benchmark is saturated.** On 150 generated queries every configuration scores above 0.958 and BM25 alone reaches 0.9951 — a benchmark where everything wins cannot rank anything. Its queries sit too close to the source text.
- **On one benchmark BM25 beats the full pipeline** (0.8929 vs 0.8773), because those queries are verbatim body sentences, exactly the case lexical matching is built for. Reported rather than hidden; it correctly bounds where the added complexity helps.
- **Single-benchmark evaluation.** All labeled results come from SciFact, so generality is unproven; NFCorpus or TREC-COVID would test it.
- **The deployed corpus is unlabeled** and is a sample rather than a mirror of arXiv — hence both the live arXiv fallback and the click/vote logging, which is the beginning of a real relevance dataset.

All figures reproduced from results/metrics.json, results/significance.json and results/latency.json at seed 42. Code: github.com/Kagzyyy/ResearchOS · Live: researchos-brown.vercel.app